

Yujie Gao

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EDUCATION

Cornell Tech, Cornell University

MEng in Operations Research and Information Engineering, Merit Scholarship

New York, NY

May 2026

University of California, Berkeley

BA in Data Science

Berkeley, CA

May 2025

PROFESSIONAL EXPERIENCE

Goodyear Tire, Data Scientist Intern

May 2024 - August 2024

- Engineered a Python/SQL ETL pipeline in Snowflake to ingest and clean 50,000+ monthly sales records, implementing feature-specific imputation strategies reducing reporting latency by 10 hours/month.
- Processed 10M+ vehicle registration records from CAAM and applied K-Means geospatial clustering to segment 300+ markets, identifying 5 high-yield distribution nodes.
- Developed a regularized Ridge regression model for regional demand forecasting, integrating seasonal features, and macroeconomic indicators to improve out-of-sample R^2 from 0.79 to 0.82 and reduce forecast variance by 15%.

Baidu Internet Service, Data Scientist Intern (Ads Ecosystem)

May 2023- August 2023

- Managed A/B testing frameworks for ad creative ranking, utilizing internal experimentation platforms to evaluate statistical significance (p-values) and confidence intervals for high-traffic feature rollouts.
- Investigated a 5% relative Conversion Rate (CVR) drop within a specific iOS cohort via user-path segmentation, collaborating with product engineering to deploy a targeted landing-page fix that fully restored baseline stability.
- Created automated dashboards to monitor Ads platform "North Star" metrics, translating petabyte-scale behavior logs into diagnostic KPIs (eCPM, CTR, Hide-rate) to balance monetization with user retention.

PROJECTS

Bird Audio Classification & Sound Event Detection (BirdClef) | PyTorch, ONNX, CNN, Spectrum | [Deep Dive](#)

2026

Developed an end-to-end bioacoustics deep learning pipeline to identify avian vocalizations in continuous, noisy field recordings under strict computational deployment constraints.

- Engineered a scalable audio preprocessing pipeline using TorchAudio to transform raw 1D waveforms into optimized 2D Mel-spectrograms, training CNN-based backbones for multi-label sound event detection.
- Mitigated severe long-tailed class imbalance and domain shifts by designing EDA-driven data augmentation strategies, including MixUp and CutMix, to enhance model generalization against complex background noise.
- Leveraged Knowledge Distillation to transfer logit predictions from an ensemble of deep teacher architectures into a lightweight student model, boosting classification AUC by over 2% while reducing parameter bloat.
- Exported production-ready PyTorch weights to an ONNX Runtime environment and implemented a multithreaded CPU inference pipeline, accelerating execution speeds by 10% to satisfy low-latency edge deployment requirements.

Algorithmic Pricing & Dynamic Programming Engine | XGBoost, K-Means, Backward Induction | [Deep Dive](#)

2025

Developed an end-to-end dynamic pricing system that handles real-time revenue optimization by decoupling offline machine learning demand estimation from an online finite-horizon optimization solver.

- Built a finite-horizon dynamic programming pricing solver with K-Means state discretization and precomputed policies, enabling $O(1)$ online pricing decisions under a 0.5-second inference constraint.
- Developed an XGBoost purchase-probability model in Scikit-learn to support pricing optimization, outperforming logistic regression baselines by 39.1% in total profit on a 50K-customer test set in projected revenue.
- Deployed an adaptive feedback loop to a competitive tournament, algorithmically adjusting pricing strategies based on recent transaction win-rates and inventory depletion speeds, securing a 10th place rank out of 36 teams in static evaluation.

TECHNICAL SKILLS

Languages & Infrastructure: Python, SQL, SparkSQL, Snowflake, ONNX Runtime, Multithreaded Inference, ETL Pipelines.

ML & Exp: PyTorch, XGBoost, Scikit-learn, Deep Learning (CNN), A/B Testing, Causal Inference, Statistical Modeling.